

MEMORY

Long-term memory, Recall vs recognition, is memory reliable?

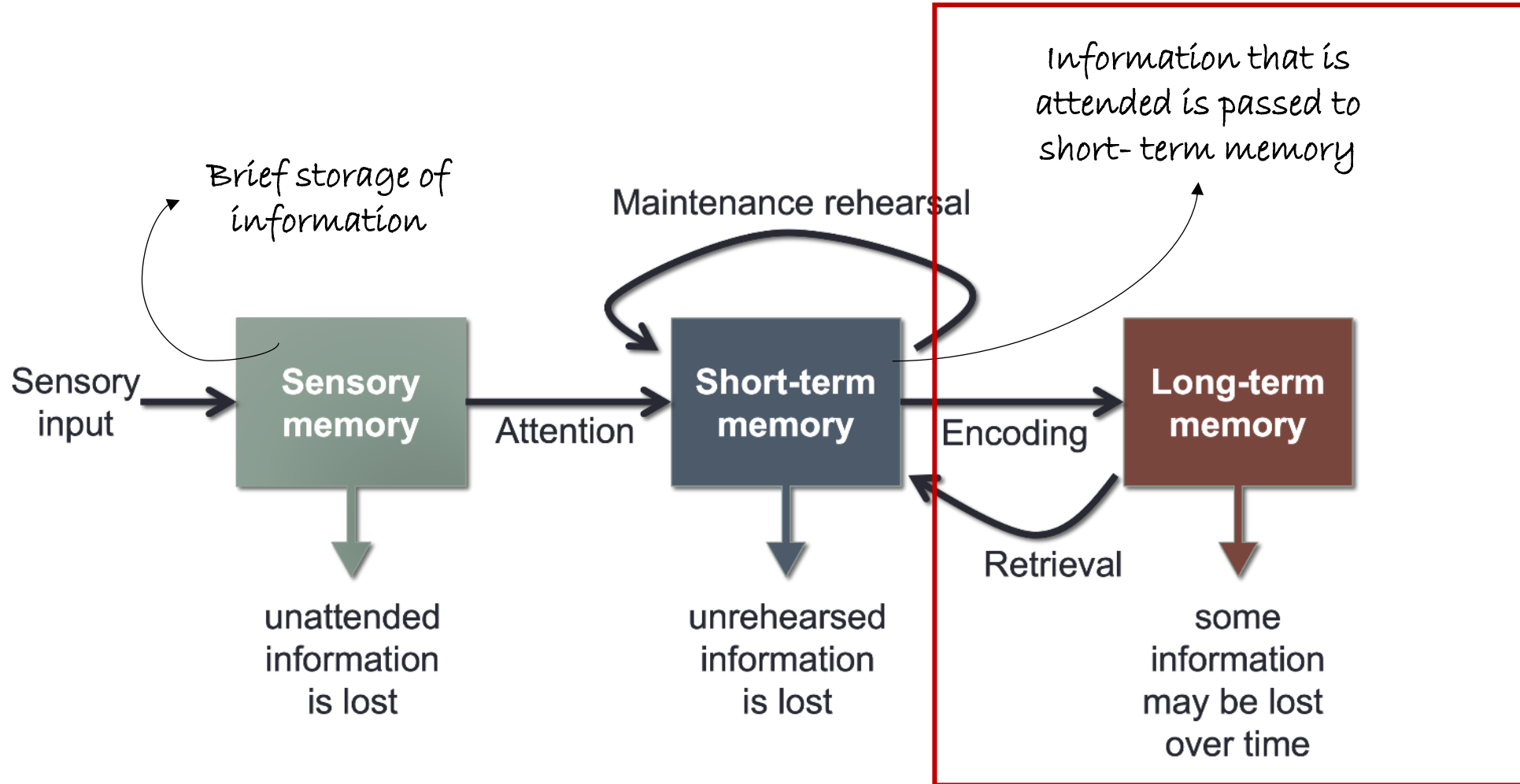
Course instructor:
Dr Shweta Chawak
Assistant Professor
Humanities and Social Sciences
IIT Bombay



RECAP

- Memory is an active system that receives information from the senses, puts that information into a usable form, organizes it as it stores it away, and then retrieves the information from storage
 - Putting it (Encoding)
 - Keeping in storage (holding information)
 - Getting it out (Retrieval)
- Information processing model
 - Sensory memory – short term memory – long-term memory
 - Sensory memory: raw sensory information hold [iconic (visual) & echoic (audio)]
 - Short term memory: information held for shorter duration (7 ± 2)
- How to hold information: maintenance rehearsal or chunking

Information processing model



ACTIVITY: MEMORY GAME!

- Remember your first day on this campus!
- Remember the words we discussed in last class?

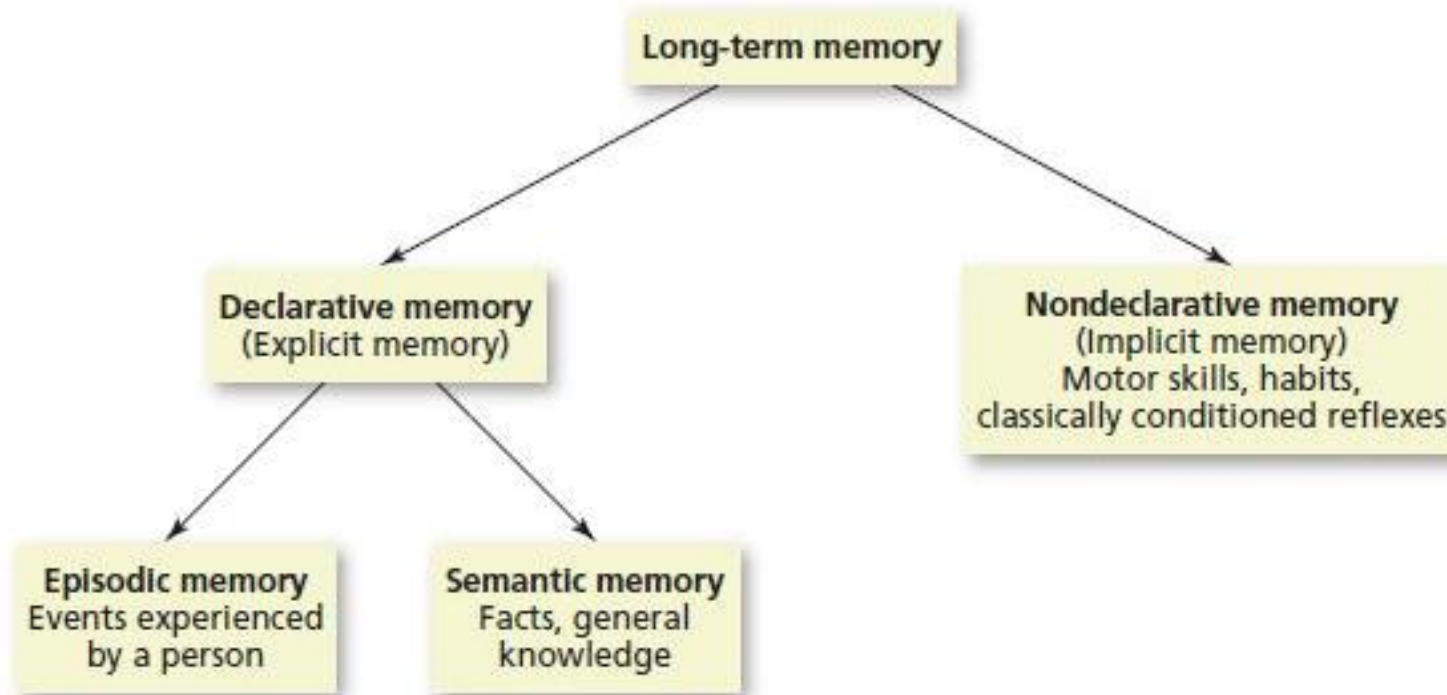
How do we store memories?! Long-term Memory

Long-term memory (LTM):

- The system of memory into which all the information is placed to be kept more or less permanently
- Encoded in terms of images, sounds, smells, or tastes
- Stored as long-lasting memories of events and concepts that are meaningful and important



How do we store memories?! Long-term Memory



In thinking about a typical day, how do you use each type of memory: nondeclarative, episodic, and semantic?

How do we store memories?! Long-term Memory

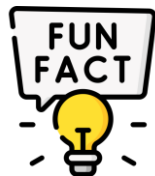
Types of Long-term Information (LTM): Declarative (Explicit) memory

- All the things that people **can know**—the facts and information that make up knowledge
- “What was for brunch?”
- Conscious and known



Semantic memory:

- The word *semantic* refers to meaning
- Awareness of the meanings of words, concepts, and terms as well as names of objects, math skills, and so on



Episodic memory:

- Memories of what has happened to people, such as certain birthdays, childhood events that were particularly special
- Represent episodes from their lives



How do we store memories?! Long-term Memory

Types of Long-term Information (LTM): Nondeclarative (Implicit) memory

- Memories for things that people know **how to do**, like tying shoes and riding a bicycle
- Emotional associations (amygdala), habits, and simple conditioned reflexes (hind brain)
- These memories are not conscious but are implied to exist because they affect conscious behaviour
- The kind of memories people “never forget”

- **Anterograde amnesia:** new long-term declarative memories cannot be formed
- Loss of memory from the point of injury or trauma forward
- Damage to the hippocampal area of the brain
- Alzheimer’s disease, who also suffer from anterograde amnesia, do not forget how to walk, talk, fasten clothing, or even tie shoes





How do we store memories?! Long-term Memory

Elaborative rehearsal

- Way of increasing the number of retrieval cues (stimuli that aid in remembering) for information by connecting new information with something that is already well known
- Information processed according to its **meaning**, not mere sounds or words
- Remembered more efficiently and for a longer period of time

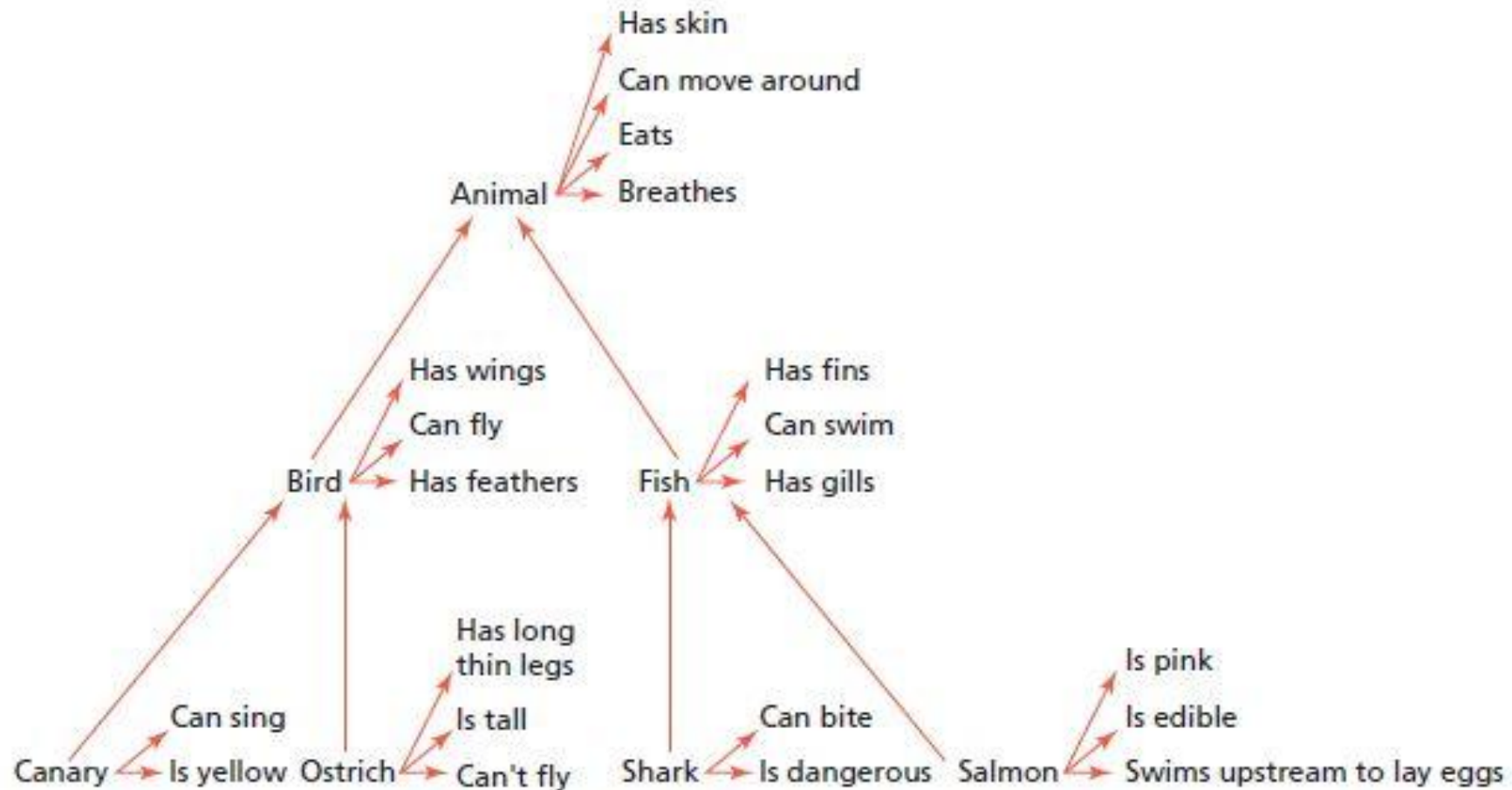


E.g.,

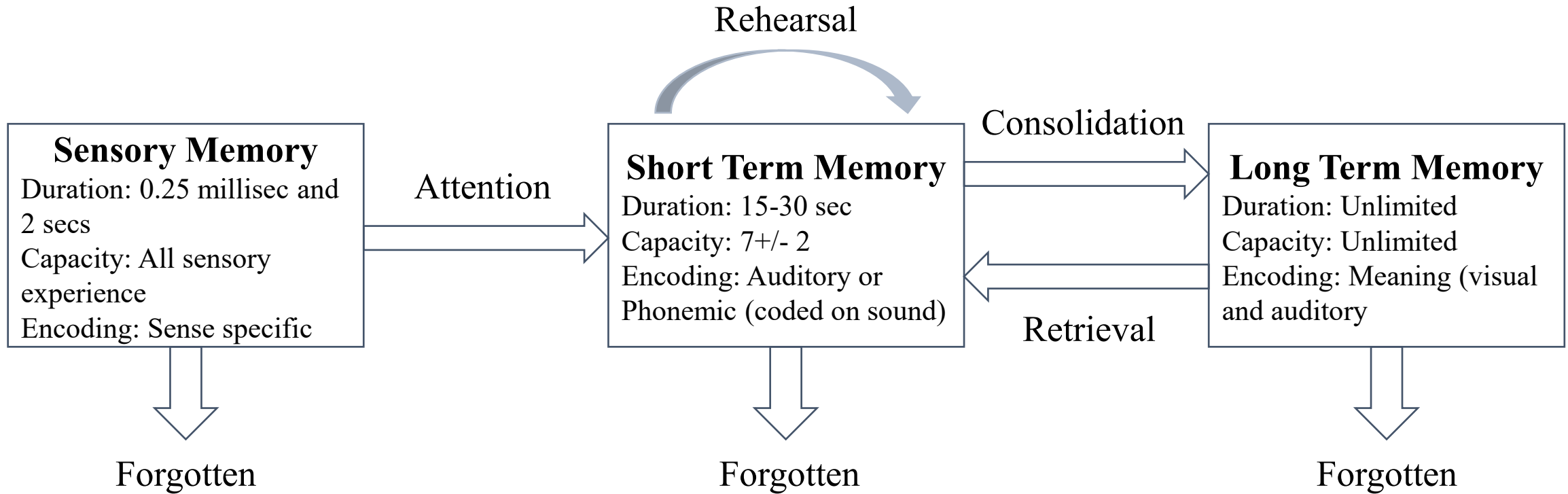
- **Personal analogy**: Connect the concept of “operant conditioning” to getting more likes on Instagram posts after posting funny memes, encouraging more meme-sharing
- **Using familiar songs**: Set the structure of the brain to the tune of a favorite pop song for easier recall during exams
- **Password creation**: Use a sentence or a familiar phrase with meaningful substitutions—“MyDogBen2019!” combines a pet’s name and the year adopted
- **Remembering someone’s name**: If introduced to a person named Lily, visualize a lily flower or connect her name to a “Lily” character from a TV show watched regularly



How do we store memories?! Long-term Memory



An Example of a Semantic Network



How do we retrieve memories? Recall versus Recognition

Recall

- Memories are retrieved with few or no external cues
- E.g., filling in the blanks on an application form

Tip of the tongue (TOT) phenomenon

Serial position effect

- Tendency of information at the **beginning and end** of a body of information to be remembered more accurately than information in the middle of the body of information

Primacy effect

- Words at the beginning of a list are remembered better
- Early words get more rehearsal time in short-term memory
- Middle words are replaced quickly by new incoming words, so they are less rehearsed

Recency effect

- The last words on a list are remembered easily
- These words are still in short-term memory (STM)
- No new words come in to push them out, making retrieval easier

Are multiple-choice tests easier than essay tests? If so, why?

How do we retrieve memories? Recall versus Recognition

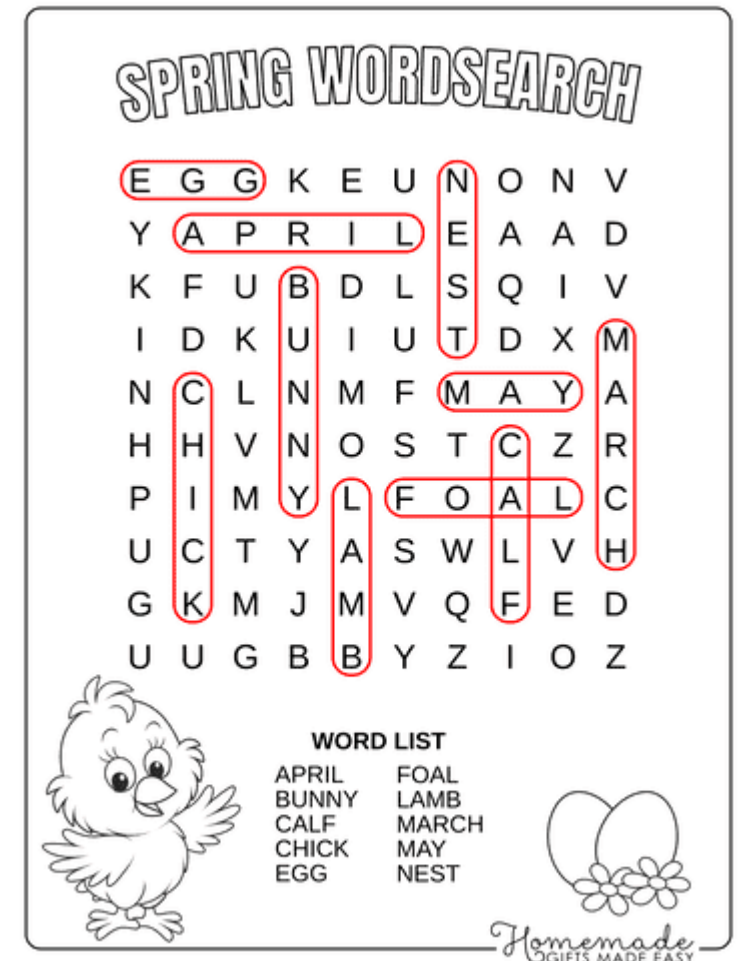
If you're given a list of 12 words: "table, chair, window, door, spoon, lamp, book, pen, apple, orange, milk, water," you might recall:

- **Primacy Effect:** Words at the beginning, like "table," "chair," and "window"
- **Recency Effect:** Words at the end, like "milk" and "water"

How do we retrieve memories? Recall versus Recognition

Recognition

- Involves looking at or hearing information and **matching** it to what is already in memory
- E.g., word-search puzzle and multiple-choice
- Recognition is accurate for images, especially human faces
- Susceptible to false positive: incorrectly recognize something as correct



How reliable are our memories?

Hello.

How reliable are our memories? (cont'd...)

Constructive processing of memory:

- Memories are **built or reconstructed** from stored information, not simply retrieved as exact copies
- Each time a memory is recalled, it can be changed; new information may be added or some details left out, depending on what is available at the time of reconstruction
- E.g., after learning new details about an event, people often revise their memories to match what they now know, discarding previous mistakes or inaccurate information
- “*Hindsight bias*” where people mistakenly believe they “knew it all along” after learning the actual outcome
- E.g., after receiving a graded test, a student might say, “I knew I should have studied that topic more,” even though before the test, they actually were uncertain or didn’t think it was important



Why do we forget?

Concept	Definition	Example
Encoding Failure 🚫	When information is never processed into long-term memory	Not remembering someone's name because attention was on the phone during introductions
Storage Failure 📉	Loss of information from memory over time due to decay or non-use	Forgetting details of last semester's lectures because never reviewed or used them
Retrieval Failure ❓	Inability to access stored information due to missing cues	Experiencing "tip-of-the-tongue" when trying to recall a familiar password or a celebrity's name
Interference 🔄	When other memories block retrieval <u>Proactive interference</u>: older information already in memory interferes with the learning of newer information <u>Retroactive interference</u>: newer information interferes with the retrieval of older information	<u>Proactive interference</u>: Remembering an old address makes it difficult to recall a newly moved-to address <u>Retroactive interference</u>: Learning a new phone number makes it harder to remember your old number
Transfer of Memory 🔄	Applying or failing to apply learned information across contexts/situations	Struggling to use memory strategies from psychology class when memorizing a grocery list for the first time

Retroactive affects the *past* (old memories), proactive affects the *future* (new memories)

Memory Techniques

- ☐ Recitation: repeating information verbally until it is encoded into long term memory
- ☐ Rehearsal: silently repeating information until it is encoded into long term memory
- ☐ Selection: highlighting important information instead of reading an entire page
- ☐ Organization (Chunking): separating information into smaller chunks of information to remember
- ☐ Spaced practice: studying a small amount of information for a short time, then taking a break before studying more
- ☐ Mnemonic: any kind of memory system or aid (e.g., James-Lange theory = Jumpy Legs)



RECAP

- The system of memory into which all the information is placed to be kept more or less permanently
- Information processed according to its **meaning**, not mere sounds or words
- Remembered more efficiently and for a longer period of time
- Types of LTM:
 - Declarative:
 - Semantic
 - Episodic
 - Non-declarative
- Elaborate rehearsal – Retrieval cues – Semantic network
- Recall vs Recognition
- Constructive process of memory
- Why do we forget?
- Memory techniques

References

- Ciccarelli, S. K., White, J. N., & Misra, G. (2017). *Psychology* (5th ed.). Pearson Education India.

Gifs, memes, images, supplementary reading resources, & icons taken from:

- <https://www.flaticon.com>